A LOCAL RING MAP WITH FINITE RESIDUE EXTENSION ADMITS A PRIME OUTSIDE THE CLOSED FIBRE WHOSE QUOTIENT IS QUASI-FINITE OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 14957-14967 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 14969-14991, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. First of the chapter's quasi-section pair; the sibling card (nearby-open-X variant) sits further down the queue and is a different statement.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Jacobian-ness of the punctured spectrum of a local Noetherian ring (Properties, lemma-complement-closed-point-Jacobson), the isolated-point fibre criterion for quasi-finiteness (Algebra, lemma-isolated-point-fibre), and openness of the quasi-finite locus (Algebra, lemma-quasi-finite-open).

Source reference. Original lemma 00PK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14991–14991. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00QA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14991–14991. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02IM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14991–14991. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\varphi : A \to B$ be a local ring map of local rings. Let $V \subset \mathrm{Spec}(B)$ be an open subscheme which contains at least one prime not lying over $\mathfrak{m}_A$. Assume A is Noetherian, φ essentially of finite type, and $A/\mathfrak{m}_A \subset B/\mathfrak{m}_B$ is finite. Then there exists a $\mathfrak{q} \in V$, $\mathfrak{m}_A \neq \mathfrak{q} \cap A$ such that $A \to B/\mathfrak{q}$ is the localization of a quasi-finite ring map.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Since A is Noetherian and $A \to B$ is essentially of finite type, we know that B is Noetherian too. By Properties, Lemma 02IM the topological space $\mathrm{Spec}(B) \setminus \{\mathfrak{m}_B\}$ is Jacobson. Hence we can choose a closed point $\mathfrak{q}$ which is contained in the nonempty open

$$V \setminus \{\mathfrak{q} \subset B \mid \mathfrak{m}_A = \mathfrak{q} \cap A\}.$$

(Nonempty by assumption, open because $\{\mathfrak{m}_A\}$ is a closed subset of $\mathrm{Spec}(A)$.) Then $\mathrm{Spec}(B/\mathfrak{q})$ has two points, namely $\mathfrak{m}_B$ and $\mathfrak{q}$ and $\mathfrak{q}$ does not lie over $\mathfrak{m}_A$. Write $B/\mathfrak{q} = C_{\mathfrak{m}}$ for some finite type A-algebra C and prime ideal $\mathfrak{m}$. Then $A \to C$ is quasi-finite at $\mathfrak{m}$ by Algebra, Lemma 00PK (2). Hence by Algebra, Lemma 00QA we see that after replacing C by a principal localization the ring map $A \to C$ is quasi-finite. $\square$